Internet Crimes and Cyber Security

Week: 06 **Session: 29**

Topic : **Basics of Cyber Forensics**

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Basics of Cyber Forensics

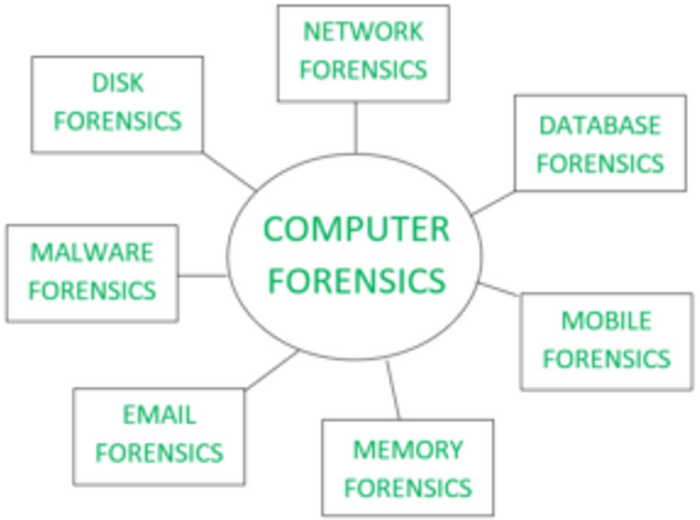
Concepts to be covered

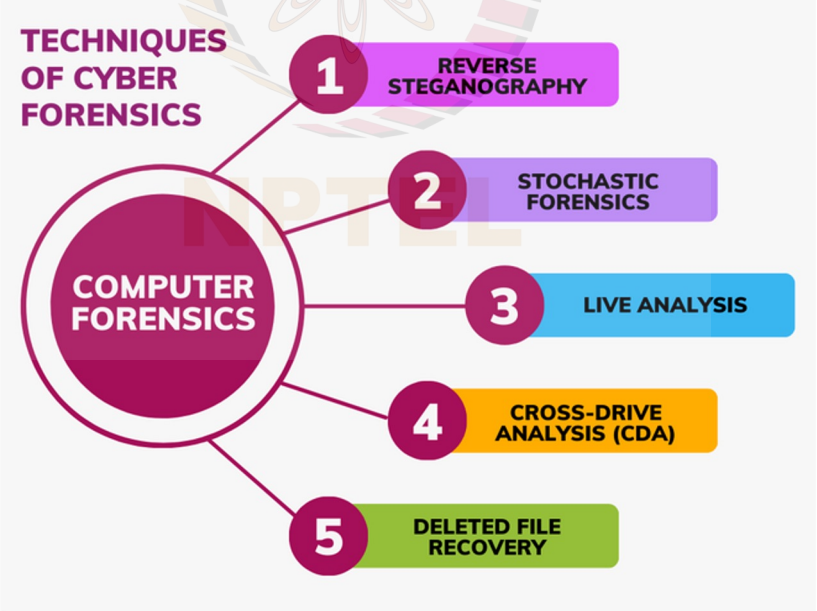
1. What is Cyber Forensics
2. What is Digital Evidence
3. Exploring the characteristics and types of Digital Evidence
4. Understanding Cyber Forensics Life Cycle
5. Tools and Techniques for Cyber Crimes Investigation
6. Live Case Study : Corporate Data Breach
7. Conclusion and Way Ahead

Basics of Cyber Forensics

1. What is Cyber Forensics

Cyber forensics involves the collection, preservation, analysis, and presentation of digital evidence related to cybercrimes. This field is crucial for investigating cyberattacks, data breaches, and other illegal activities conducted in the digital realm. The primary goal is to uncover how an attack occurred, identify the perpetrators, and gather evidence that can be used in legal proceedings.



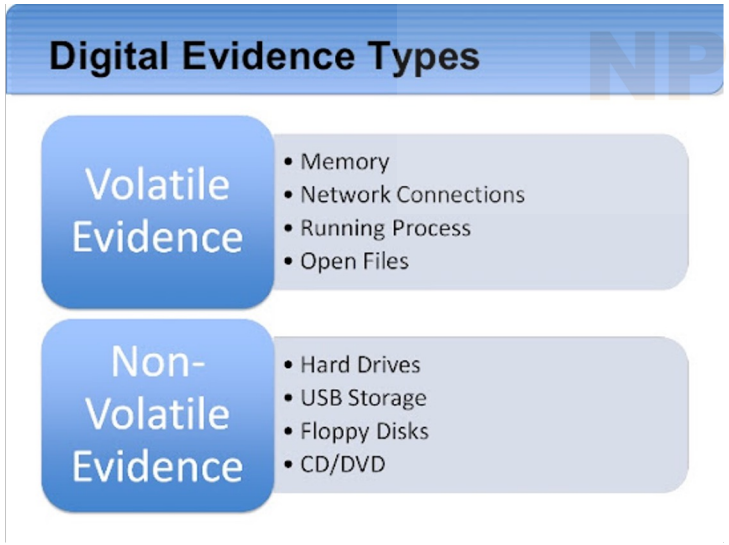


2. What is Digital Evidence

Digital evidence encompasses a wide range of information derived from electronic devices, including computers, smartphones, and cloud storage.

It plays a crucial role in modern legal practices, as it can substantiate claims during investigations and trials.

Examples of digital evidence include emails, GPS records, chat messages, and metadata, which can provide definitive timelines and insights into events or interactions



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3. Exploring the characteristics of Digital Evidence

Latent: Digital evidence is often not immediately visible or accessible. It can be hidden within devices or files, similar to how fingerprints are latent until revealed through proper techniques.

Fragile: Digital evidence is highly susceptible to alteration, damage, or destruction. Even minor changes in the environment or handling can compromise its integrity, making careful collection and preservation essential.

Time-Sensitive: The relevance and availability of digital evidence can diminish over time. Data can be deleted, overwritten, or corrupted, necessitating prompt action to secure it.

Intangible: Unlike physical evidence, digital evidence exists in a binary format and is often stored on electronic devices such as computers, smartphones, and servers. This intangibility can complicate its collection and presentation in legal contexts.

Complexity: Digital evidence can be complex due to the variety of formats and sources it can originate from, including emails, text messages, multimedia files, and internet browsing history. This complexity requires specialized knowledge for effective analysis and interpretation.

Transnational: Digital evidence can easily transcend national borders, which can complicate legal proceedings and the enforcement of laws across jurisdictions.

Computer Fraud Investigations:

- Account data from online auctions
- Accounting software and files
- Address books
- Calendar
- Chat Logs
- Customer information
- Credit card data
- Databases
- Digital camera software
- E-mail, notes and letters
- Financial and asset records

Child Abuse and Pornography Investigations:

- Chat logs
- Digital camera software
- E-mails, notes and letters
- Games
- Graphic editing and viewing software
- Images
- Internet activity logs
- Movie files
- User created directory and file names which classify images

Network Intrusion Investigations:

- Address books
- Configuration files
- E-mails, notes and letters
- Executable programs
- Internet activity logs
- Internet protocol address & usernames
- Internet relay chat logs
- Source code
- Text files and documents with usernames and passwords

Homicide Investigations:

- Address books
- E-mails, notes and letters
- Financial asset records
- Internet activity logs
- Legal documents and wills
- Medical records
- Telephone records
- Diaries
- Maps
- Photos of victim / suspect
- Trophy photos

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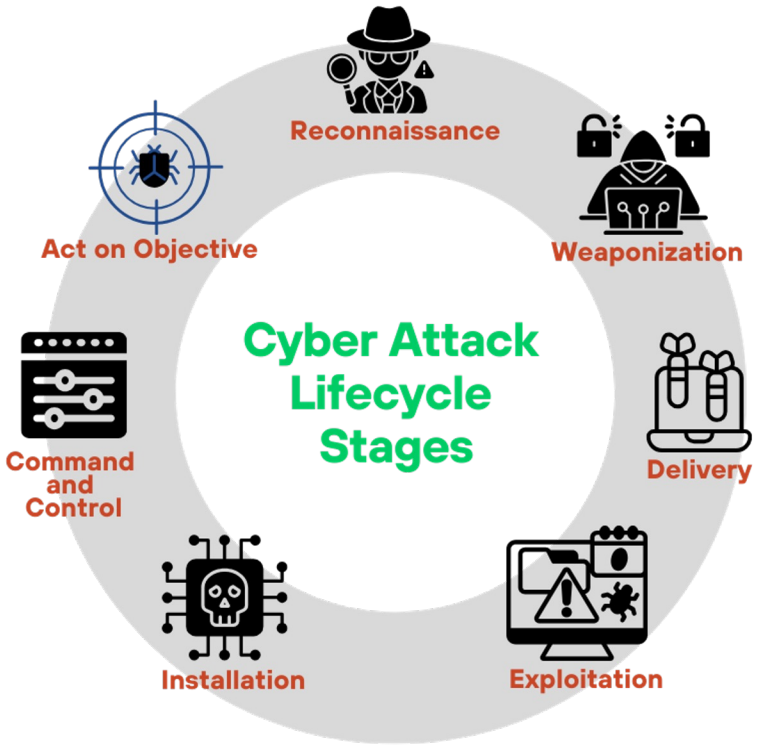
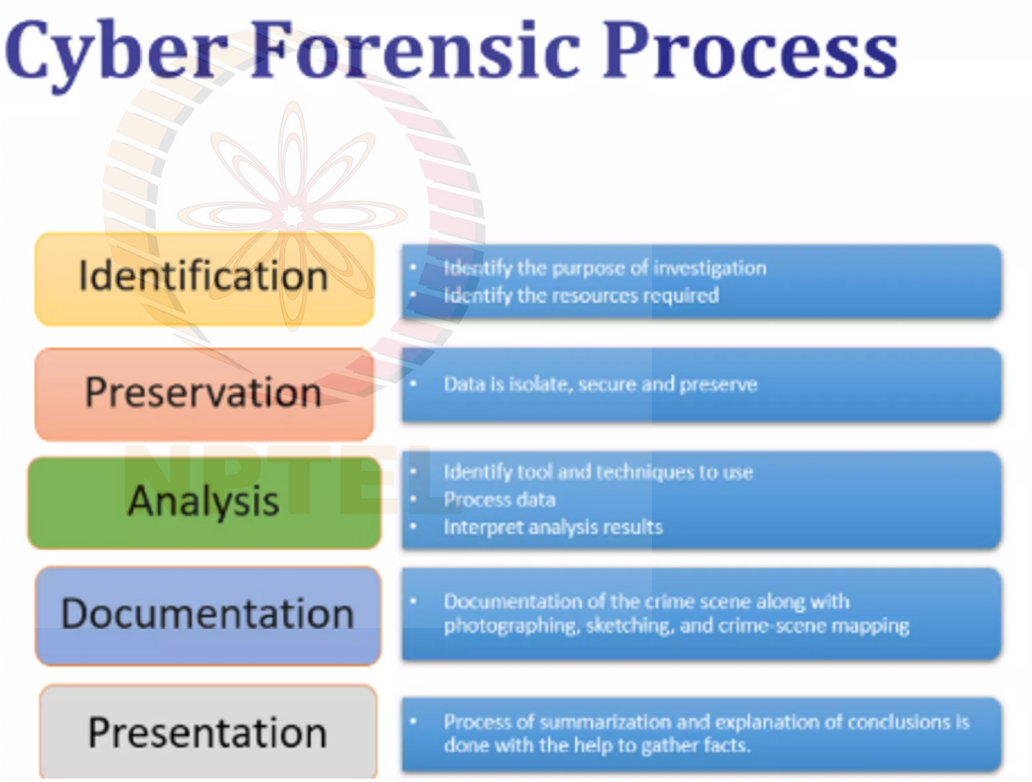
4. Understanding Cyber Forensics Life Cycle





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5. Tools and Techniques for Cyber Crimes Investigation

AUTOPSY
DIGITAL FORENSICS

Open-Source and Specialized Tools

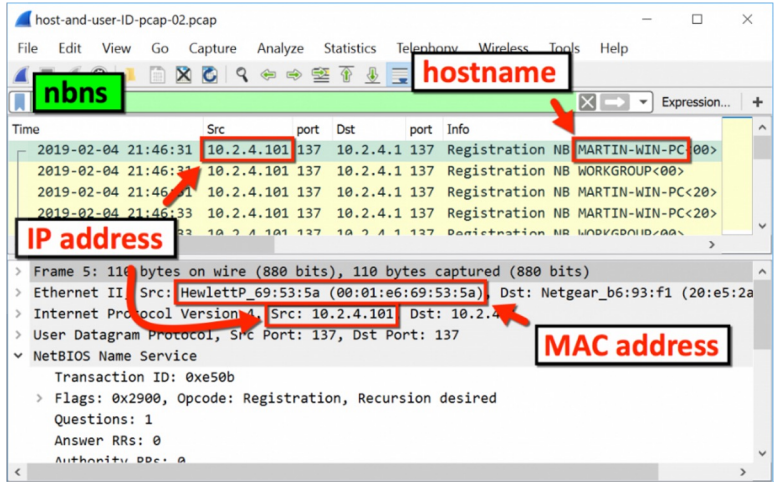
1. Autopsy/The Sleuth Kit (TSK) – Open-source platform with a GUI for disk imaging, file recovery, timeline reconstruction, and plugin support for social media or cloud artifacts. Ideal for beginners, academics, or budget-conscious investigations

2. Volatility Framework – A memory forensics tool for analyzing RAM images, detecting malware, rootkits, open processes, and user activity. Supports multiple OS memory formats and is frequently updated by a community of researchers

3. X-Ways Forensics – Efficient and resource-light tool for disk imaging, file recovery, and detailed analysis of file systems. Suitable for forensic professionals who need deep insights without heavy system demands

4. SANS SIFT Workstation / Paladin Forensic Suite – Pre-configured Linux-based collections of open-source forensic utilities including Sleuth Kit, Volatility, and imaging tools. Useful for quick deployment, incident response, and triage in diverse environments

5. Wireshark – Though primarily a network protocol analyzer, it is an important tool for forensic investigations, offering packet inspection and analysis for network breaches, data exfiltration, and intrusion detection



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6. Live Case Study : Corporate Data Breach

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7. Conclusion and Way Ahead

The future of digital forensics is poised for significant advancements, driven by the need to address the evolving landscape of cybercrimes. Here are some key points to consider for the future of digital forensics:

Emerging Technologies: The integration of AI, machine learning, and blockchain is expected to revolutionize forensic practices, making it easier to analyze complex data and track cryptocurrency transactions.

Legal and Ethical Challenges: As digital evidence becomes more prevalent, the legal and ethical implications of data privacy regulations and trans-jurisdictional data management will continue to be critical considerations.

Standardization and Collaboration: The need for standardization of forensic techniques and ethical rules governing AI use will be essential for maintaining the integrity of digital forensics.

Cross-Jurisdictional Cooperation: Collaborating across borders to combat shifting cyber threats will be crucial for the global community to effectively address digital forensics challenges.

Thank You